PREDICTING CUSTOMER CHURN IN E-COMMERCE

Leveraging Statistical Models and Hypothesis Testing for Strategic Insights

Abstract

This project explores factors influencing customer churn in e-commerce using statistical models and synthetic data. Despite applying EDA, hypothesis testing, and logistic regression with oversampling to balance classes, the model's poor accuracy (~50%) highlights the limitations of synthetic data and the need for real-world datasets for actionable insights.

Research Question

What are the key factors that predict customer churn in an e-commerce environment, and how can statistical models, including hypothesis testing and regression analysis, be used to identify these factors and optimize marketing strategies?

Data Collection

- The data for this project was sourced from Kaggle.
- It is a synthetic dataset created artificially using the Faker Python library.
- The dataset mimics real e-commerce data, including customer behavior and purchase history in an online marketplace.

https://www.kaggle.com/datasets/shriyashjagtap/e-commerce-customer-for-behavior-analysis/data?select=ecommerce_customer_data_large.csv

Data Overview

Customer Age: Numerical

Gender: Categorical

Product Category: Categorical

Product Price: Numerical

Quantity: Numerical

Total Purchase Amount: Numerical

Payment Method: Categorical

Returns: Categorical (binary)

Data Preprocessing

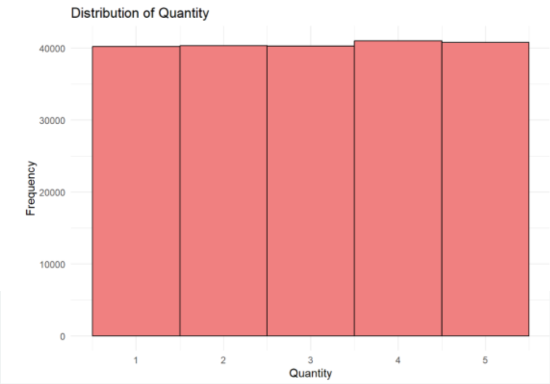
Handling Missing Values: Removing the rows with missing values in the dataset

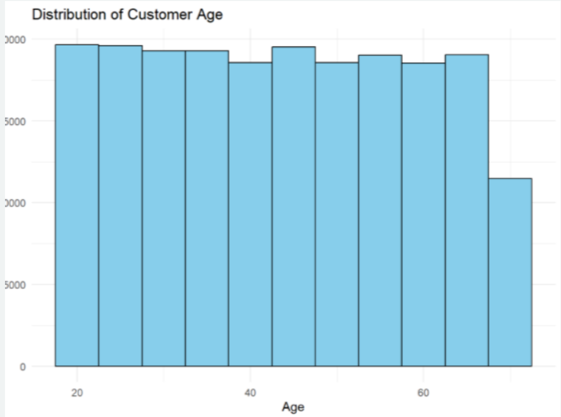
Data Cleaning: Updating Total Purchase Amount Column

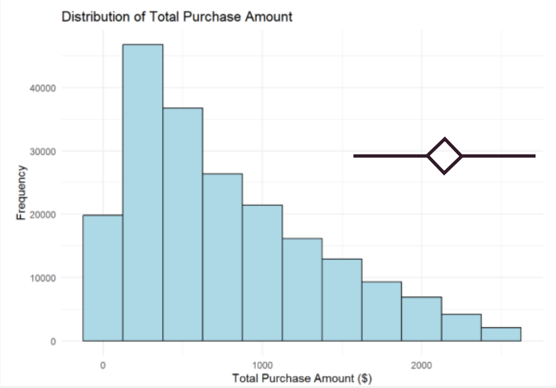
The Total Purchase Amount column in the dataset was found to have incorrect values. This issue was resolved by recalculating the column as the product of Product Price and Quantity. The updated column was generated within the R environment without modifying the original dataset on GitHub.

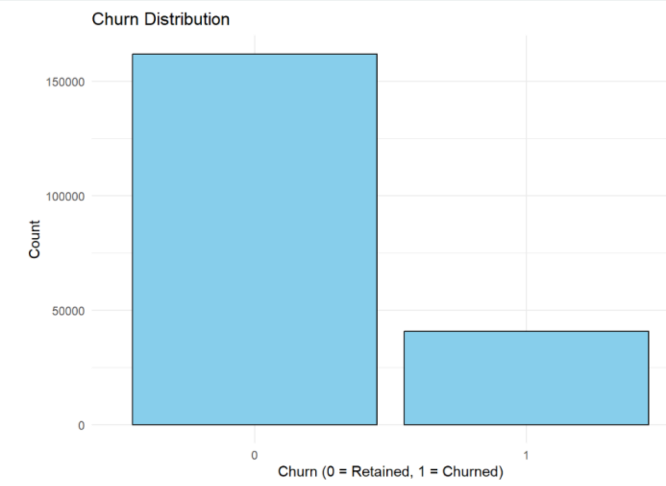
EXPLORATORY DATA ANALYSIS

UNIVARIATE ANALYSIS - NUMERICAL VARIABLE





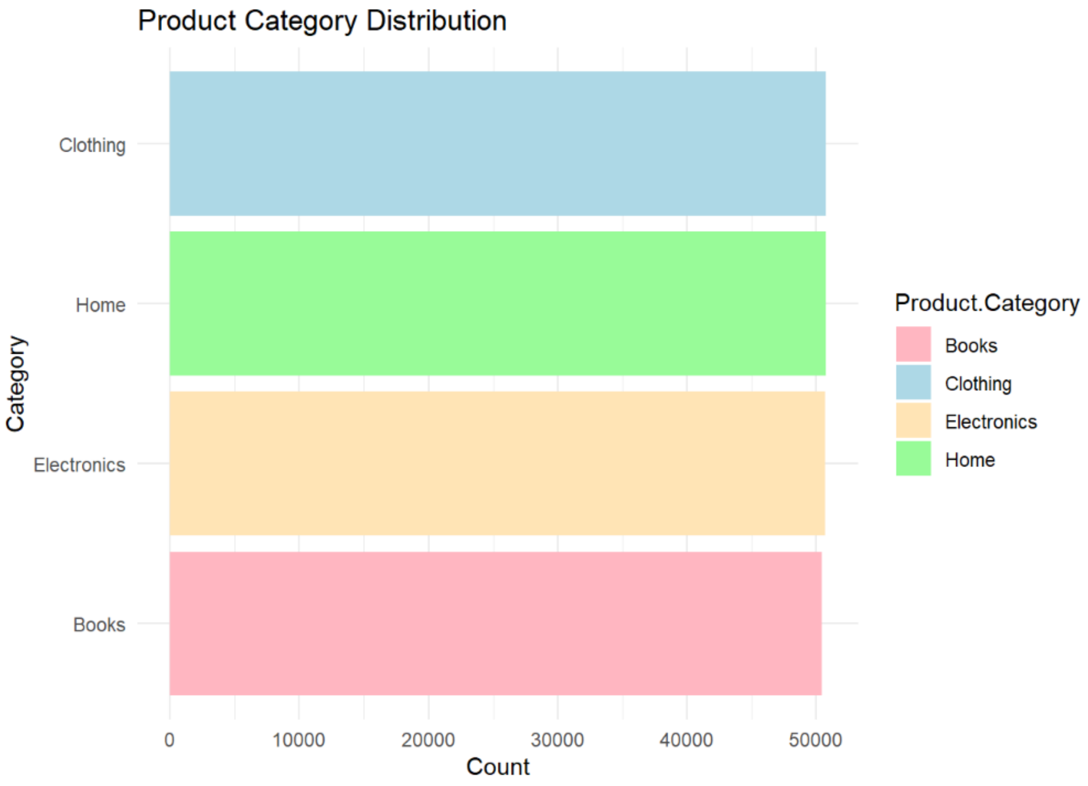




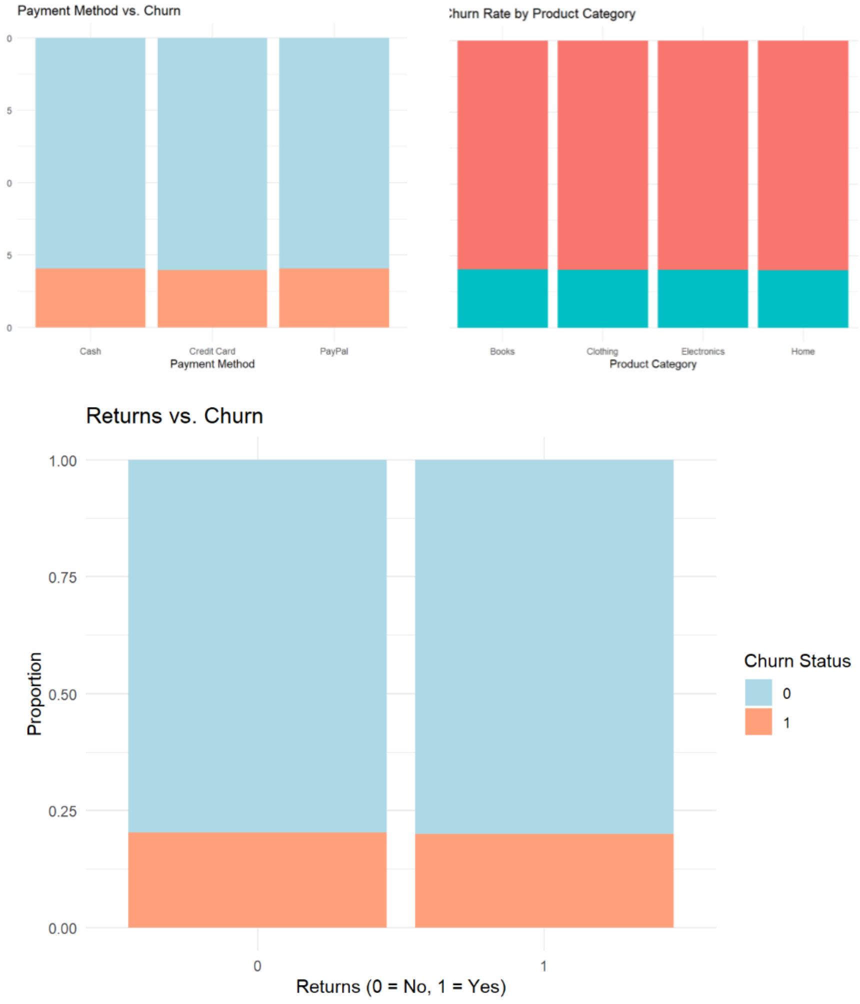
Univariate Analysis – Categorical Variable

Gender	
Female	100699
Male	101919

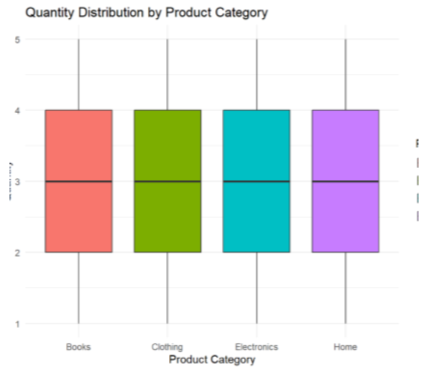
Payment Method	
Cash	67290
Credit Card	67517
PayPal	67811

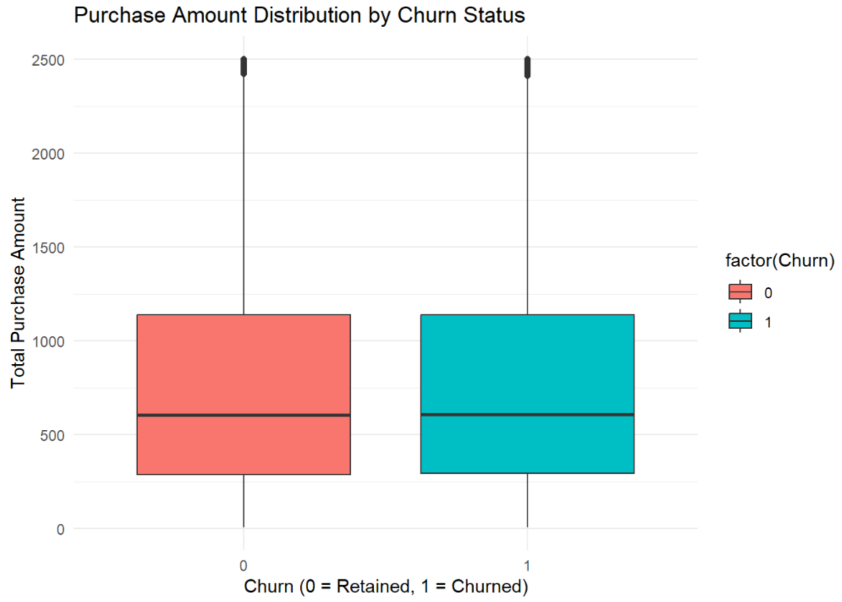


BIVARIATE ANALYSIS :BAR PLOTS

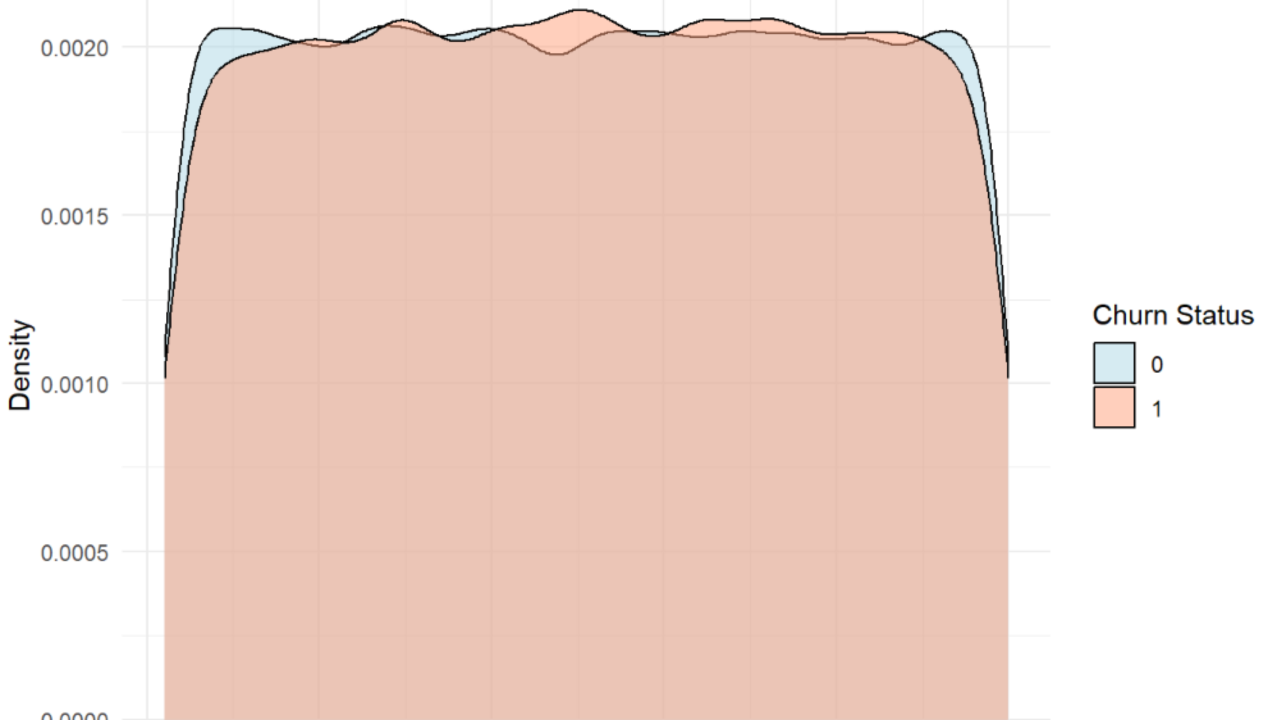


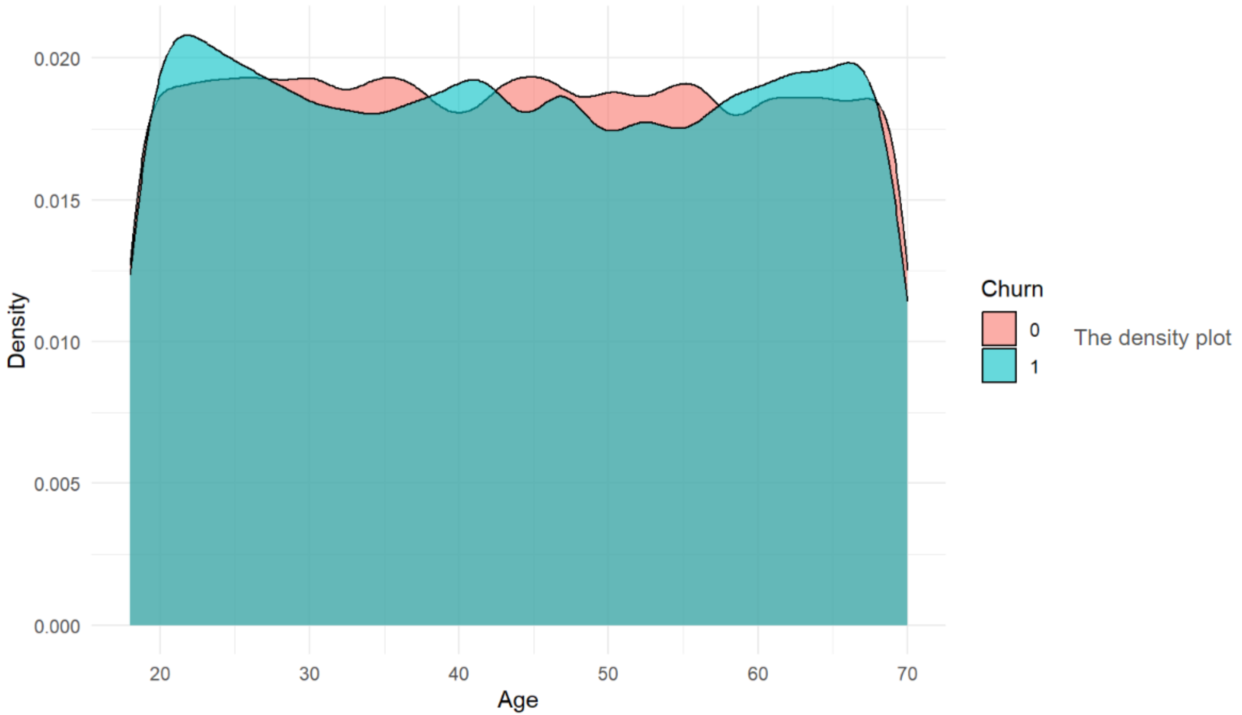
BIVARIATE ANALYSIS :BOX PLOTS



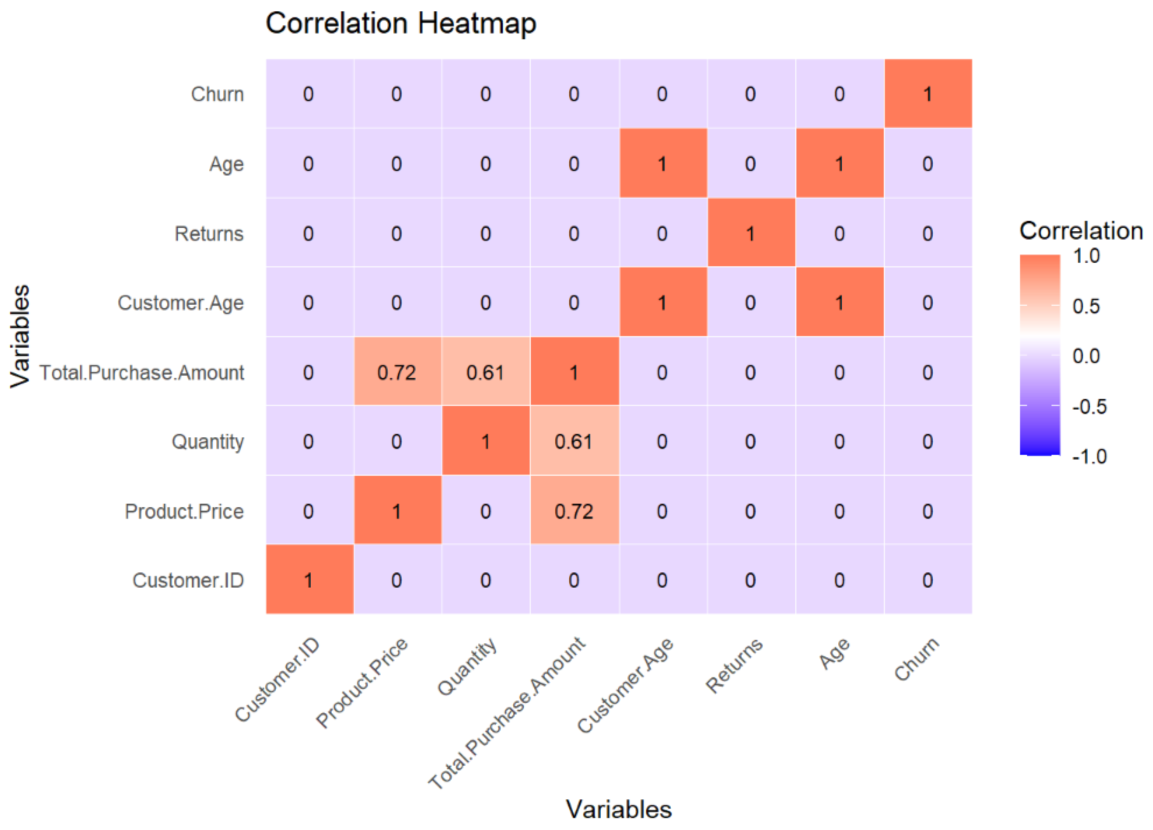


BIVARIATE ANALYSIS :DENSITY PLOTS

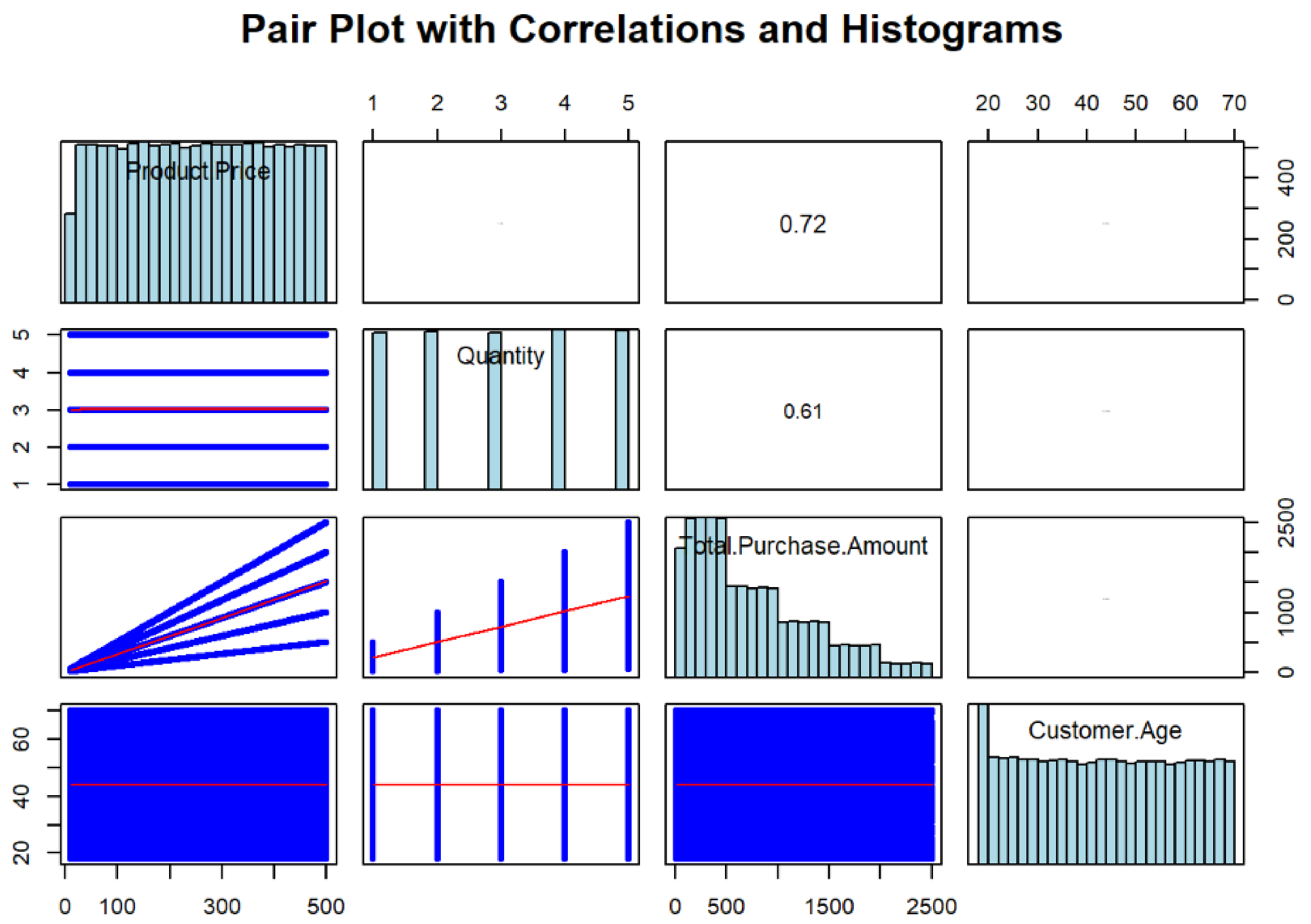




Correlation Heatmap



Multivariate Analysis



Statistical Tests: Evaluating Relationships and Differences

Chi- Squared Test : Categorical Variables and Churn

Objective: To identify significant relationships between variables and customer churn to support model-building and insights.

Null Hypothesis (H0): Category variable does not influence churn.

Alternative Hypothesis (H1): Category variable influences churn.

- **Chi-Squared Test:** Used to test associations between categorical variables (e.g., Payment Method, Returns) and churn.
- Findings: Payment Method is the only categorical variable showing a significant relationship with churn. This indicates it could be an important factor to consider.
- Product.Category, Returns, and Gender do not show strong relationships with churn based on this analysis.

T- Test: Numerical Variables and Churn

- **T-Test:** Compared means of numerical variables (e.g., Product Price, Customer Age) across churned and non-churned groups.

Null Hypothesis (H_0): The means of the numerical variable are the same across the two Churn groups.

Alternative Hypothesis (H_1): The means of the numerical variable are significantly different across the two Churn group

- Findings: Minimal differences were observed between groups, indicating weak predictive power.
- None of the numerical variables showed a statistically significant difference in means across churned and retained groups. This suggests that these numerical variables might not be strong predictors of churn based on mean comparison alone.
- Purpose: To validate assumptions about feature relevance before incorporating them into the logistic regression model.

Statistical Test Results Summary

Test	Variable(s)	Test Type	p-value	Significance
Chi-Squared Test	Payment Method	Categorical	0.01203	Significant (p < 0.05)
Chi-Squared Test	Returns	Categorical	0.06725	Not Significant (p > 0.05)
Chi-Squared Test	Gender	Categorical	0.222	Not Significant (p > 0.05)
Chi-Squared Test	Product Category	Categorical	0.8463	Not Significant (p > 0.05)
T-Test	Product Price	Numerical (mean)	0.2334	Not Significant (p > 0.05)
T-Test	Quantity	Numerical (mean)	0.3073	Not Significant (p > 0.05)
T-Test	Total Purchase Amount	Numerical (mean)	0.3646	Not Significant (p > 0.05)
T-Test	Customer Age	Numerical (mean)	0.3336	Not Significant (p > 0.05)

Logistic Regression Model

Objective: To model the likelihood of customer churn based on key predictors.

Independent Variables:

- Product Price
- Quantity
- Total Purchase Amount
- Customer Age
- Returns

Process: Logistic regression was applied to predict churn probability.

- Accuracy of the initial model: ~50%.
- Significant predictors included Quantity and Returns.

Findings: The model's low predictive power highlights weak relationships between features and churn.

Synthetic data characteristics may contribute to limited model performance.

Tackling Class Imbalance for Model Improvement

Challenge: Churn data is imbalanced: ~80% non-churned, ~20% churned.

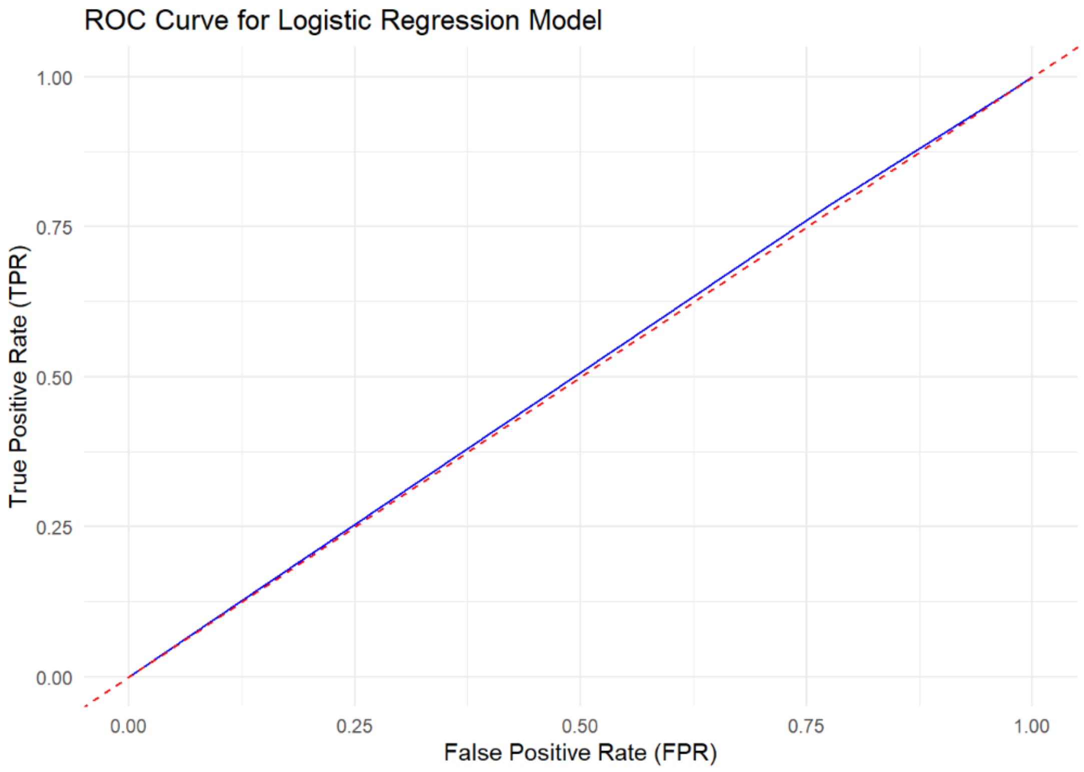
Imbalance can bias the model toward predicting the majority class.

Solution: Oversampling of the minority class to equalize churned and non-churned instances.

- Created a balanced dataset with a 50:50 ratio.
- **Result:** Accuracy remained ~50% post-balancing, indicating weak feature-churn relationships.
- **Implication:** Class imbalance alone does not explain low model performance, Data quality and feature relevance are critical factors.

Model Evaluation

Without Stratified Training



With Stratified Training

LIMITATIONS

1. **Synthetic Data:**

- The data was artificially generated and did not reflect the complexity of real-world e-commerce environments.
- Key drivers of churn, such as customer satisfaction and engagement, were not included.
- This limitation prevented the model from finding meaningful patterns for churn prediction.

2. **Class Imbalance:**

- Although oversampling and stratified sampling addressed class imbalance, the lack of strong predictors limited the impact on model performance.

3. **Limited Feature Relevance:**

- Many features in the dataset, like Product Category and Payment Method, did not significantly correlate with churn.

Important features, like customer behavior or interaction history, were missing.

4. **Results Not Generalizable:**

- Findings from this study are specific to the synthetic dataset and cannot be applied to real-world e-commerce scenarios without further validation.

CONCLUSION

1. No Strong Predictors of Churn:

- Features like Product Price, Customer Age, Product Category, and Gender showed little to no influence on predicting churn.
- Only Returns and Quantity showed slight significance but were not strong enough to improve predictions.

2. Model Performance:

- The logistic regression model achieved an accuracy of ~50%, which is similar to random guessing.
- The ROC AUC score of ~0.5 further confirmed the model's inability to differentiate between churned and non-churned customers.

3. Methodology Provides a Framework:

- Despite the limitations, the applied methods, such as hypothesis testing, stratified sampling, and logistic regression, provide a foundation for future analysis with better data.

THANK YOU